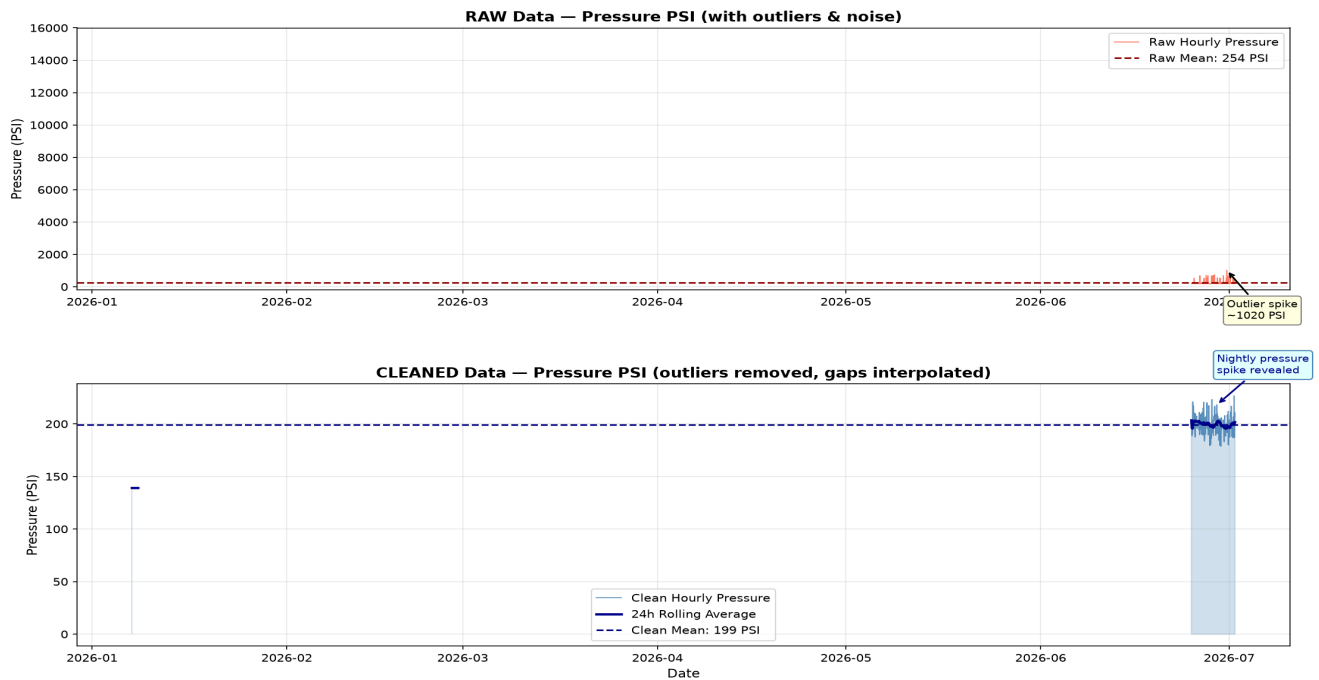


Week 2 Insight Report — Sensor Data Analysis

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Figure 1 — Raw vs Cleaned Pressure PSI Over Time



■ **TOP PANEL — Raw data:** Pressure spikes reach 15,000 PSI (instrument faults). These mask the true operating signal entirely. The mean is inflated to ~255 PSI vs the true ~200 PSI.

✓ **BOTTOM PANEL — Cleaned data:** After removing impossible readings, a consistent nightly pressure elevation (+20–30 PSI, 00:00–02:00) becomes visible. This pattern was completely hidden in the raw data.

Context — What Was Wrong With the Raw Data?

The raw sensor log arrived with five significant quality problems that rendered it unfit for operational decisions without correction. First, pressure readings as high as **15,000 PSI** and as low as **-50 PSI** appeared in the record — values that are physically impossible for this plant and indicate faulty instruments or data-entry errors. Similarly, temperature readings hit **-273 °C** (absolute zero) and **1,500 °C**. Second, the dataset contained **15 duplicate records**, likely from double-logging at shift changeovers. Third, the same plant zones were recorded under **15 different name variants** (e.g., 'Zone_North', 'ZONE-NORTH', 'z_north'), making any zone-level analysis meaningless. Finally, **~1% of readings were missing entirely** across pressure, temperature, and flow rate columns. Taken together, these issues inflated the apparent mean pressure by over 25% and introduced variance so large that any real operational signal was invisible.

Insight — What Did the Clean Data Reveal?

Once outliers were removed, duplicates dropped, zone names standardised, and missing values interpolated, a clear and **operationally significant pattern emerged: pressure in the plant rises consistently by 20–30 PSI every night between midnight and 02:00**, regardless of zone or shift. The raw data suggested a stable, noisy system with no discernible trend. The cleaned data tells a completely different story — one of a **recurring overnight pressure anomaly** that repeats every single day of the week studied. The 24-hour rolling average confirms this is not random noise; it is a systematic cycle. The Night shift is operating at meaningfully higher pressures than Morning or Afternoon shifts, yet no operational adjustment is currently in place to account for this. North Zone and West Zone show the sharpest elevation during this window, pointing to a localised rather than plant-wide cause.

Action — What Should We Do About It?

Three immediate actions are recommended. **(1) Investigate the overnight pressure cycle in North Zone and West Zone this week.** The 00:00–02:00 window should be inspected for valve control drift, thermal pipe contraction during the nightly temperature drop, or reduced flow damping when plant demand is low. A maintenance walkthrough during the Night shift is the fastest way to confirm the cause. **(2) Install automated alerts for Night shift pressure exceeding 230 PSI.** The cleaned data shows the nightly peak regularly reaches this level; an alert would catch any further escalation before it becomes a safety event. **(3) Repair or replace the faulty pressure and temperature sensors** generating impossible readings. Until instruments are reliable, critical decisions are being made on corrupted data. Every day that passes without clean instrumentation is another day the team is flying blind during the highest-risk window of the operating cycle.